1) The conf. space is: $\underbrace{SO(2)}_{\phi} \times \underbrace{\mathbb{R}}_{\ell} \times \underbrace{SO(2)}_{\theta}$, is a 3-manifold.

The constraint is:

$$\underbrace{\begin{pmatrix} m(d+\ell)^2 & 0 & J \end{pmatrix}}_{A} \begin{pmatrix} \dot\phi \\ \dot\ell \\ \dot\theta \end{pmatrix} = 0 \quad \text{hence a basis for } \mathcal{N}(A) \text{ is:}$$

$g_2 = (0 \;\; 1 \;\; 0)^T \qquad g_1 = (1 \;\; 0 \;\; \alpha)^T$ where:

$$Ag_2 = m(d+\ell)^2 + J\alpha = 0 \Rightarrow \alpha = -\frac{1}{J}m(d+\ell)^2 \quad \text{hence the model is:}$$

$$\begin{cases} \dot\phi = \omega \\ \dot\ell = v \\ \dot\theta = -\frac{1}{J}m(d+\ell)^2\omega = \beta(d+\ell)^2\omega \end{cases} \qquad \beta := -\frac{1}{J}m$$

ϕ, ℓ are actuated, hence ω, v are the angular and linear velocity of the joints.

Local mobility is not complete, ϕ and θ cannot have two arbitrarily independent velocities. Let's consider the accessibility distribution Δ_A.

$$\text{let } g_3 = [g_1, g_2] = \underbrace{\frac{\partial g_2}{\partial q} g_1}_{0} - \frac{\partial g_1}{\partial q} g_2 = -\beta(2\ell + 2d)\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

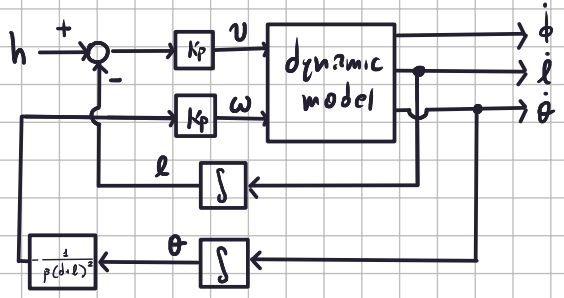
Is linearly independent from g_2, hence $\operatorname{rank} \Delta_A = 3 \Rightarrow$ global mobility

$$\begin{cases} \dot\phi = \omega \\ \dot\ell = v \\ \dot\theta = \beta(d+\ell)^2\omega \end{cases}$$

to drive (ℓ, θ) to $(0, h)$ we use the following control inputs:

$$\begin{cases} v = K_p(h - \ell) & \leftarrow \text{linear control} \\ \omega = -(\beta(d+\ell)^2)^{-1} K_p \theta & \leftarrow \text{Feedback lin.} \end{cases}$$

under this law $\Rightarrow \begin{cases} \dot\ell = K_p(h-\ell) \\ \dot\theta = -K_p\theta \end{cases}$ hence they converge exponentially.



Now we set $d = 0 \Rightarrow \begin{cases} \dot\phi = \omega \\ \dot\ell = v \\ \dot\theta = \beta\ell^2\omega \end{cases}$

ϕ, θ are flat outputs, and $\dot\theta = \beta\ell^2\omega \Rightarrow \ell = \sqrt{\frac{\dot\theta}{\beta\omega}} = \sqrt{\frac{\dot\theta}{\beta\dot\phi}}$ Assumption $\ell \geq 0$!

The reconstruction is:

$$\begin{cases} v = \Psi_1(\dot\theta) = \dot\theta \\ \omega = \Psi_2(\dot\phi) = \dot\phi \\ \ell = \Psi_3(\dot\theta, \dot\phi) = \sqrt{\dot\theta/\beta\dot\phi} \end{cases}$$

Let $s \in [0,1]$ the path variable.

$\ell = \left(\frac{\theta'}{\beta \phi'}\right)^{1/2}$ so i set

$\theta'(0) = \ell_s^2 \quad \theta'(1) = \ell_g^2 \quad \phi'(0) = 1/\beta \quad \phi'(1) = 1/\beta$

hence $\theta(s)$, $\phi(s)$ must be quadratic (2 boundary condition).

$\theta = as^3 + bs^2 + cs + d$, let's find the coefficients

$$\begin{cases} \theta(1) = a+b+\ell_s^2+\theta_s = \theta_g \\ \theta(0) = d = \theta_s \\ \theta'(1) = 3a+2b+\ell_s^2 = \ell_g^2 \\ \theta'(0) = c = \ell_s^2 \end{cases}$$

we solve for a, b and d the same for $\phi(s)$.

---

2) At each time step, ϕ_k and ℓ_k are known from the encoders, these are exactly what we impose trough ω, v. At k-th step, we have an estimation of the conf. that is $N(\hat{q}_k, P_k)$, a normal distribution. We can have a first estimation of $\tilde{q}_{k+1}$ by using Euler integration:

$$\begin{cases} \tilde{\phi}_{k+1|k} = \hat{\phi}_k + T_s \omega \\ \hat{\ell}_{k+1|k} = \hat{\ell}_k + T_s v \\ \hat{\theta}_{k+1|k} = \hat{\theta}_k + T_s \beta (d+\hat{\ell}_k)^2 \omega \end{cases} = f(\hat{q}_k) = \hat{q}_{k+1|k}$$

$\{\omega, v$ reconstructed via encoders

by assuming there is a noise $N(0, V_k)$ on the dynamics, we compute a first estimate of P_{k+1} as $P_{k+1|k} = F_k P_k F_k^\top + V_k$ where $F_k = \frac{\partial f}{\partial q}\Big|_{q=\hat{q}_k, u=u_k}$

$$F_k = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & T_s\beta(2\hat{\ell}_k + 2d)\omega & 1 \end{pmatrix}$$